GAME DESIGN DOCUMENT (GDD): KAUDI

Document Version: 1.0 (Production Master)
Project: Native & Indigenous Sports of India — Title 18
Institution: Centurion University of Technology and Management (CUTM)
Date: September 2026

PART 1: CONCEPT DOCUMENT

Title Page

- **Game Name**: Kaudi: The Ancient Cowrie Shell Scoring Game
- **Game Logo**: Stylized golden cowrie shell (*Cypraea moneta*) framed by circular temple mandala engravings.
- **Game Catch Phrase**: *"Cast the shells, master the odds, revive the sacred game."*
- **Document Type**: Full Game Design Document (GDD) & Technical Specification
- **Document Version**: 1.0 (Gold Master Specification)

Credit Page

- **Document Purpose**: Authoritative game design and technical blueprint for the digital adaptation of the traditional Indian cowrie-shell dice and scoring sport.
- **Working Title**: *Project Kaudi Core (IKS-Title-18)*
- **Game Concept**: Digital preservation and gamification of ancient Indian cowrie-shell casting, functioning both as a standalone competitive scoring game and as the shared deterministic dice engine for the entire CUTM Indian Board Games Suite.
- **Author / Studio**: Centurion University Heritage Game Development Studio (Lead: Swarup & Core Team)
- **Lead Designer**: Abhi Mitra (Design Head, CUTM)
- **Academic Reference**: *An Anthology of Native & Indigenous Sports of India* (CUTM, 2025)

Sign-Off

Role	Name	Status	Signature Date
Lead Artist	Heritage Art Coordinator	Approved	2026-09-29
Lead Designer	Abhi Mitra	Approved	2026-09-29
Lead Programmer	Core Systems Engineer	Approved	2026-09-29
Lead Producer	CUTM Academic & Project Director	Approved	2026-09-29

Introduction

Kaudi is a digital adaptation of traditional Indian cowrie-shell throwing games played across the Indian subcontinent for over three millennia. Natural cowrie shells serve as both dice and counters: players toss them onto a woven mat and score points based on how many shells land aperture-up (mouth open) versus dorsal-up (flat back).

- **Genre**: Casual Digital Board Game / Traditional Strategy & Probability Race.
- **Player Type**: Casual gamers, families, folklore enthusiasts, and students (all ages: 6 to 80).
- **Technical Form**: High-performance 2D Canvas / WebGL with lightweight physics simulation.
- **References**: Excavations at Mohenjo-Daro and Harappa; classical Sanskrit treatises (*Mahabharata* dice references, *Chaturanga* preludes); *CUTM Indigenous Sports Anthology* (2025).
- **Theme**: Indian Antiquity, Rustic Courtyard, Tactile Heritage Craftsmanship.
- **Design Intentions**: Cultural preservation through playable authenticity; authentic hand-made material feel; zero-dependency fast web loading.

Game Analysis

Game Description

Kaudi is a turn-based multiplayer and single-player scoring competition. Players cast a clutch of polished cowrie shells (4, 6, or 7) into a framed wooden rolling tray lined with natural coir

fabric. Points are awarded based on exact mouth-up counts, with special milestone configurations (all shells mouth-up, or all shells mouth-down) offering high scores and bonus turns. Players compete to reach a predefined target score (typically 100 points).

Genre & Elements

- **Genre**: Traditional Board / Dice / Push-Your-Luck Casual.
- **Core Game Elements**:
  - *Tossing*: Shaking and releasing shells into the play arena.
  - *Reading / Scoring*: Immediate visual identification of mouth-up shells.
  - *Push-Your-Luck Decision*: In Risk mode, deciding whether to bank accumulated round points or risk a bust roll to accelerate toward victory.
  - *Tactical Banking*: Safeguarding scores against aggressive opponent pacing.
- **Game Content**: Wholesome, historical, family-friendly, educational, culturally enriching.

Theme & Style

- **Theme**: Heritage Indian Courtyard, temple sandstone, hand-carved teakwood, woven coir, brass oil lamps.
- **Style**: Warm, tactile, skeuomorphic-modern. Real physical textures with crisp digital UI readability.
- **Game Sequence**: Turn-based, linear rounds with dynamic push-your-luck sub-sequences.
- **Player Count**: 1 to 4 players (1P vs AI, 2P–4P local pass-and-play).

Game Reference & Taxonomy

- **Taxonomy**: Ludology-dominant game of chance and risk assessment, grounded in authentic Indian ethnographic history.
- **Caillois Classification**: *Alea* (Chance/Probability) with elements of *Agon* (Competitive race) and *Ilinx* (Excitement of press-your-luck stakes).
- **Player Immersion**:
  - *Tactile Immersion*: Organic shell rattle, clatter, bounce, and settling dynamics.
  - *Strategic / Probability Immersion*: Calculating risk vs. reward on each subsequent throw.
  - *Cultural Immersion*: Authentic terminology (*Chamma*, *Ashtam*), traditional Indian motifs, and historical background lore.

Game Technical

- **Technical Form**: 2D Orthographic / Top-Down View with layered pseudo-3D parallax and drop-shadows.
- **Camera View**: Fixed top-down perspective focusing on the central wooden rolling mat with player status cards flanking the periphery.
- **Target Platforms**:
  - Primary: WebGL / Modern Web Browsers (Chrome, Edge, Safari, Firefox) + PWA (Progressive Web App).
  - Secondary: Mobile Android (.aab, API Level 36 compliant) and iOS (iOS 16+ via Xcode).
- **Languages & Stacks**: Plain C# (Unity 6.3 LTS engine core) / Vanilla JavaScript (HTML5 Canvas web distribution).
- **Target Devices**: Low-to-high-end smartphones, tablets, smart school boards, and desktop PCs.

Game Sales & Distribution

- **Consumer Group**: Broad demographic spanning K-12 students, universities, diaspora communities, casual mobile gamers, and board game preservationists.
- **Monetization Model**: Free-to-play educational showcase with zero pay-to-win mechanics. Optional non-intrusive cosmetic unlocks (custom shell varieties and wood tray finishes).
- **Estimated Price**: 100% Free Open-Access as part of the Centurion University Cultural Heritage Initiative.

Game Atmosphere

The atmosphere captures the peaceful golden hour on a traditional Indian verandah (*verandah / dhalani*). The ambient soundscape includes distant temple bells, gentle evening wind, and soft acoustic classical ragas played on Bansuri and Sitar. The centerpiece is the wooden dice tray: aged teakwood with beveled borders and a coir mat that responds with crisp, satisfying acoustic clicks whenever shells impact the surface.

Gameplay Walkthrough

1. **Match Initialization**: The player launches the game, selects player count (e.g., 2-Player Local or Player vs. AI), chooses the shell configuration (default 6 shells), and sets target points (default 100).

2. **The Turn**: Player 1 taps the "Toss Shells" action or presses Spacebar. The shells lift, rotate through simulated angular momentum, scatter across the tray, and clatter to a halt.

3. **Evaluation**:
    - If 4 of 6 shells land mouth-up, the game announces: "4 Mouths Up! +4 Points."
    - If 0 shells land mouth-up (*Ashtam*), the game triggers celebration particles: "All Down! ASHTAM! +12 Points & Bonus Roll!"

4. **Push-Your-Luck (Risk Mode)**: The player is offered two choices:
    - **Bank**: Lock the current round's +12 points into permanent score and hand turn to Player 2.
    - **Roll Again**: Cast the shells again to accumulate more points; however, if a bust throw occurs, all unbanked points from this turn are wiped out.

5. **Winning Condition**: The first player whose banked score reaches or exceeds 100 points triggers the victory sequence with celebratory confetti, conch shell horn stings, and match summary statistics.

Key Features & Selling Features

- **Dual Play Modes**: Standard Classic Scoring vs. Adrenaline-fueled Press-Your-Luck.
- **Master Reusable Engine**: Deterministic `CowrieDice` logic powering the entire 18-title Indian games suite.
- **Authentic Physics & Foley**: Multi-sample real cowrie acoustic sounds recorded on hardwood and cloth.
- **Smart Heuristic AI**: Three distinct AI personalities with mathematically calculated risk thresholds.
- **Complete Offline Functionality**: Instant PWA caching enables play in rural and offline classrooms.

PART 2: DESIGN DOCUMENT

Design Version & Guidelines

- **Version**: v1.0.0 (Adherence to CUTM Design Bible v1.0).
- **Design Guidelines**:
  - All colors restricted to the master heritage palette: Teak Wood (`#5A3919`, `#8A5A2B`), Sandstone (`#FBF4E8`, `#DCC7A3`), Charcoal Ink (`#17120E`, `#3B2F26`).
  - Typography strictly limited to **Tiro Devanagari Hindi** (titles) and **Mukta** (body, buttons, stats).
  - Minimum touch target size is strictly **88 x 88 px** on mobile viewports.
  - Every player token must feature an unambiguous unique side glyph badge—never rely on color alone.

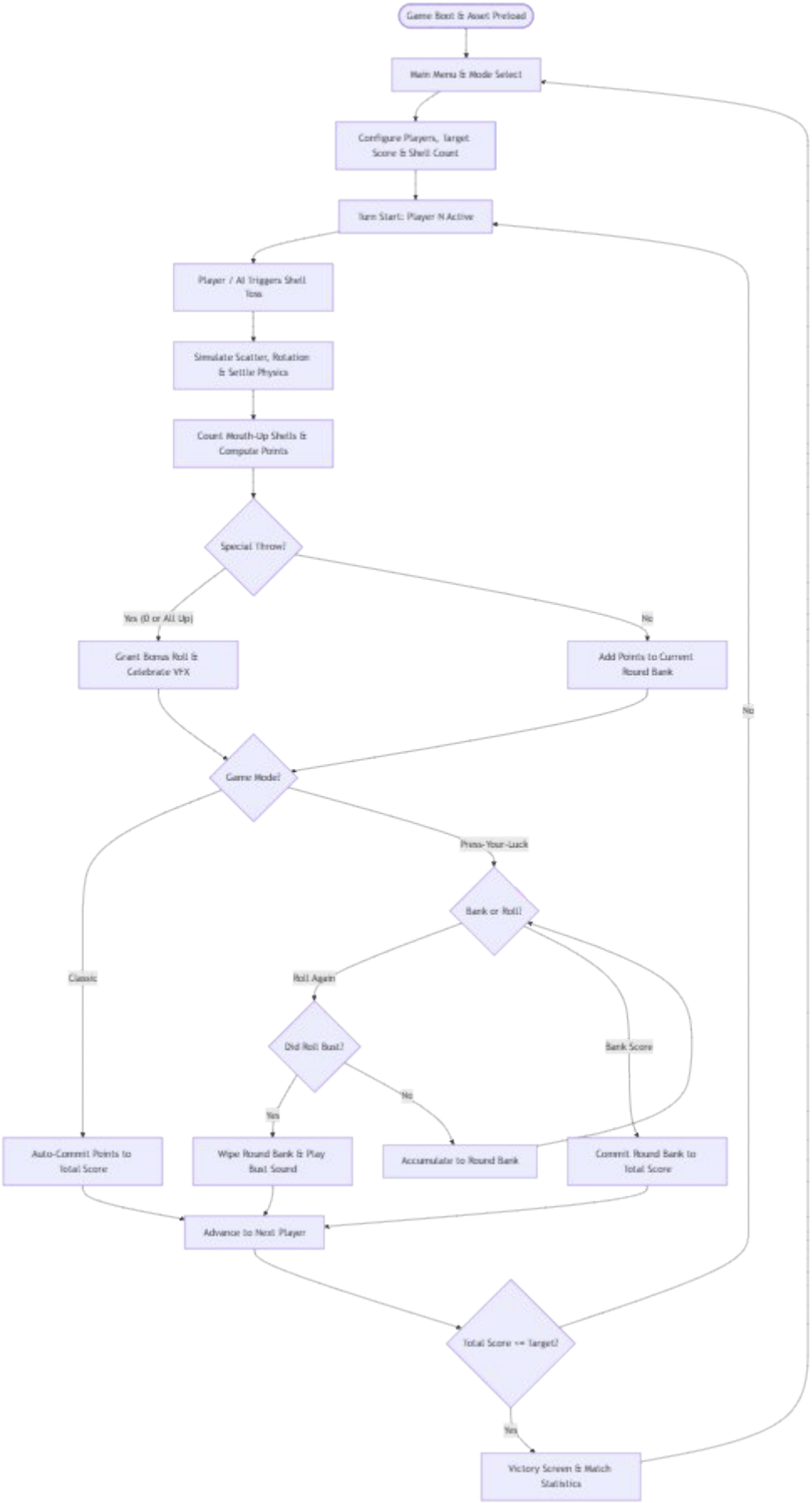
Game Design Definitions & Scoring Matrix

6-Cowrie Standard Configuration (Default)

Mouths Up	Visual Configuration	Points Awarded	Special / Bonus Throw
0	All shells face down (convex back)	**12 pts**	**ASHTAM! Bonus Roll Granted**
1	1 mouth up, 5 mouth down	**1 pt**	Normal
2	2 mouths up, 4 mouth down	**2 pts**	Normal
3	3 mouths up, 3 mouth down	**3 pts**	Normal
4	4 mouths up, 2 mouth down	**4 pts**	Normal
5	5 mouths up, 1 mouth down	**5 pts**	Normal
6	All shells face up (apertures visible)	**24 pts**	**CHAMMA! Bonus Roll Granted**

In Press-Your-Luck Mode: A roll of 1 or 0 (configurable) after an initial throw counts as a **Bust**, resetting unbanked round score to 0 and ending the turn. Consecutive bonus throws are capped at **3** to prevent infinite turn loops.

Game Flow Chart


Game Boot & Asset Preload
Main Menu & Mode Select
Configure Players, Target Score & Shell Count
Turn Start: Player N Active
Player / AI Triggers Shell Toss
Simulate Scatter, Rotation & Settle Physics
Count Mouth-Up Shells & Compute Points
Special Throw?
Yes (0 or All Up)
No
Grant Bonus Roll & Celebrate VFX
Add Points to Current Round Bank
Game Mode?
Press-Your-Luck
Bank or Roll?
Classic
Roll Again
Bank Score
Did Roll Bust?
Yes
No
Auto-Commit Points to Total Score
Wipe Round Bank & Play Bust Sound
Accumulate to Round Bank
Commit Round Bank to Total Score
Advance to Next Player
Total Score >= Target?
No
Yes
Victory Screen & Match Statistics

Player Elements & User Interface

Player Properties

- `PlayerID`: Numerical identifier (0 to 3).
- `DisplayName`: Name string ("Player 1", "Chanakya AI").
- `AvatarGlyph`: SVG/Vector badge (Lotus, Elephant, Peacock, Chakra).
- `TotalScore`: Integer tally of permanently banked points.
- `RoundBank`: Integer tally of unbanked points during the active turn.
- `BonusStreak`: Counter tracking consecutive special rolls (capped at 3).
- `IsAI`: Boolean flag indicating computer control.

Heads-Up Display (HUD)

- **Top Bar**: Game title, current round number, settings gear, and audio toggle.
- **Scoreboard Area**: 2 to 4 responsive cards showing player avatar, active turn indicator halo, numerical score, and progress bar toward target.
- **Rolling Arena**: Central wooden tray featuring physics-rendered cowrie shells, score result readout banner, and particle emitter anchor.
- **Action Footer**: Two prominent primary controls:
  - `TOSS / ROLL` Button (Teak Gold gradient, min 88px touch zone).
  - `BANK SCORE` Button (Deep Teak wood with sandstone border).

Antagonistic Elements (Artificial Intelligence)

The game features an algorithmic AI agent named **"Chanakya"** with three operational states:

1. **Normal State**: The AI waits for its turn, highlighting its avatar with a subtle brass glow.
2. **Detection State**: Upon rolling, the AI examines:
   - Current unbanked round score (B).
   - Distance to victory ($Target - TotalScore$).
   - Opponent lead / lag delta.

3. **Reaction State (Decision Policy)**:

   ○ Calculates Expected Value:

$$EV = \sum_{k \in \text{Safe}} \quad \sum_{k \in \text{Bust}}$$

   ○ If $EV > 0$ and $B < \text{RiskThreshold}$, AI triggers another toss.

   ○ If $B \geq$ RiskThreshold or if current bank wins the match immediately, AI selects **Bank**.

4. **End State**: AI finishes action, logs performance, and transfers turn cleanly to the human player.

PART 3: TECHNICAL SPECIFICATION

System Requirements

- **Browser Compatibility**: Chrome 90+, Safari 15+, Edge 90+, Firefox 88+ on desktop and mobile.
- **Memory Footprint**: < 60 MB runtime RAM usage; initial download bundle < 3 MB (uncompressed), < 1 MB (Brotli compressed).
- **Frame Rate Target**: 60 FPS continuous on mid-tier mobile hardware (Snapdragon 680 / Apple A12 or equivalent).
- **Storage**: Browser `localStorage` / IndexedDB for match history, sound settings, and achievements.

Code Structure & Class Hierarchy

```
    kaudi/
    ├── index.html              # Main WebGL / Canvas responsive
wrapper
    ├── style.css               # CUTM Design Bible CSS tokens,
layouts & animations
    ├── audio.js                # Web Audio API procedural synthesis &
Foley player
    ├── cowrie-dice.js          # Reusable deterministic Cowrie Dice
simulation engine
    └── app.js                  # Match finite state machine, HUD
manager & AI solver
```

- `CowrieDice`: Pure rules class. Simulates individual shell biases, calculates mouth-up totals, verifies special combinations, and produces reproducible seeded output.
- `AudioManager`: Manages procedural synthesis for clicks, impacts, and stings alongside audio buffers for authentic shell samples.
- `KaudiApp`: Controls match lifecycle, manages turn rotations, updates DOM UI elements, and executes AI turn timers.

Concerns, Mitigations & Alternatives

- *Concern: High variance leading to player frustration.*
  Mitigation: Implement configurable "Grace Throws" and provide dual game modes so players who dislike risk can play the predictable Classic Target Race.
- *Concern: Audio autoplay block on mobile web browsers.*
  Mitigation: Web Audio Context is initialized in a suspended state and unlocked automatically on the player's very first tap on the Start / Roll button.